

Alpha Model: Topics by Strategy

1. Alpha Quant V5 PRO (Multifactor Breakout)

- Vertical Horizontal Filter (VHF)
- Kaufman Adaptive Moving Average (KAMA)
- Ordinary Least Squares (OLS) Linear Regression
- Statistical Standardization (Z-Scores)
- Average True Range (ATR)
- Bollinger Bands & Volatility Contraction
- Python Pandas Vectorization & Rolling Windows
- NumPy Arrays & Conditional Logic

2. Momentum V1 (Top 20 Rotation)

- 6-Month Cross-Sectional Momentum Factor
- Pandas Boolean Masking & Data Filtering
- Equal Weight Portfolio Construction
- Simple Moving Average (SMA 200) Regime Filtering
- Periodic Resampling (Monthly Rotation)

3. Momentum V2 (God Mode)

- Jegadeesh-Titman Momentum (12-Month minus 1-Month)
- Inverse Volatility Portfolio Weighting
- Monte Carlo Simulations (Gaussian Noise Injection)
- Monte Carlo Simulations (Pseudo-Random Trade Shuffling)
- SMA 100 Fast Market Regime Filtering
- Walk-Forward Backtesting Validation

4. Shared Infrastructure (All Models)

- Apache Parquet / PyArrow (Columnar Data Compression)
- Object-Oriented Programming (Live State Management)
- T+1 Execution & Advanced Slippage Modeling
- REST API Data Integration (yfinance)